

RESEARCH PROFILE

Researcher in personalized AI with experience across large language models, reinforcement learning, recommender systems, and multimodal retrieval. Published 14+ papers in SIGIR, WSDM, CIKM, ECIR, IEEE TKDE, IEEE TPAMI, and related venues. Built research systems on industrial-scale interaction data at Amazon and Tencent, with recent work spanning personalized video moments and scalable composed video retrieval.

EDUCATION

University of Glasgow, Glasgow, UK

Ph.D. in Computing Science

Oct 2021 - Dec 2025

Supervisor: Prof. Joemon M. Jose

Xi'an Jiaotong University, Xi'an, China

M.S. in Software Engineering

Sep 2018 - Jun 2021

Dalian University of Technology, Dalian, China

B.S. in Software Engineering; B.S. minor in English

Sep 2014 - Jun 2018

EXPERIENCE

Amazon, Seattle, USA

Jul 2025 - Oct 2025

Applied Scientist Intern - Personalized Video Moment Recommendation

- Built a two-stage e-commerce recommender that narrows a large product-video corpus, then identifies personalized moments through user-aware visual-temporal matching.
- Evaluated the system offline and in a within-subject study with 100 real e-commerce users across nine perceptual dimensions and behavioral-efficiency outcomes.
- Product-history personalization helped users reach relevant products 36% faster and complete shopping tasks 23% faster; perceived segment usefulness improved by 0.33 points.

University of Glasgow, Glasgow, UK

Dec 2022 - Jan 2025

Teaching Assistant - Artificial Intelligence and Machine Learning

- Led laboratory instruction, coursework support, grading, and academic mentoring for master's and senior undergraduate students.
- Taught practical foundations of machine learning and reinforcement learning and supervised project-based applications.

Tencent, Shenzhen, China

Dec 2021 - May 2022

Research Intern - Transferable Multimodal Recommendation

- Developed cold-start recommendation models from 10 million user-feedback events on the QQ Browser news platform using text and visual content features.
- Supported transfer across content domains and deployment of the resulting recommendation approach on Tencent News.

SELECTED RESEARCH PROJECTS

Scalable Composed Video Retrieval with Residual-Quantized Semantic IDs

2026

Developed a frozen EVA-CLIP plus shared Q-Former retrieval encoder with 11-stage residual quantization and beam-128 semantic-ID search. On WebVid-CoVR Test, the indexed system achieves 86.05% mean Recall with 99.02% candidate recall. At a 1M-video gallery, it is measured at 16.07x faster with 36.85x less storage than an exhaustive ReTrack multi-vector baseline; adaptive temporal evidence reaches 86.81% mean Recall.

Complementary Preservation-Modification Scoring for Zero-Shot CoVR

2026

Designed a training-free framework that decomposes a composed query into preservation and modification guides, then scores invariant content and requested changes separately at frame level. The latest manuscript reports consistent gains and new training-free state-of-the-art results across three benchmarks.

PDiT-GIM: Personalized Diffusion Transformer for Guided Item Modeling

2025

Built a two-stage latent diffusion framework for personalized item generation: a diffusion transformer captures preferences in a compressed space, while a guided decoder enforces interpretable attribute constraints.

SELECTED FIRST-AUTHOR PUBLICATIONS

Large Language Model Driven Policy Exploration for Recommender Systems

Jie Wang, Alexandros Karatzoglou, Ioannis Arapakis, Joemon M. Jose
WSDM 2025, Full Paper

Reinforcement Learning-based Recommender Systems with Large Language Models for State, Reward, and Action Modeling

Jie Wang, Alexandros Karatzoglou, Ioannis Arapakis, Joemon M. Jose
SIGIR 2024, Full Paper [\[Code\]](#)

Beyond Accuracy: Decision Transformers for Reward-Driven Multi-Objective Recommendations

Jie Wang, Alexandros Karatzoglou, Ioannis Arapakis, Joemon M. Jose, Xuri Ge
IEEE Transactions on Knowledge and Data Engineering, 2025 [\[Code\]](#)

Sparks of Surprise: Multi-Objective Recommendations with Hierarchical Decision Transformers for Diversity, Novelty, and Serendipity

Jie Wang, Alexandros Karatzoglou, Ioannis Arapakis, Xin Xin, Xuri Ge, Joemon M. Jose
CIKM 2024, Full Paper

TransRec: Learning Transferable Recommendation from Mixture-of-Modality Feedback

Jie Wang, Fajie Yuan, Mingyue Cheng, Joemon M. Jose, Chenyun Yu, Beibei Kong, Zhijin Wang, Bo Hu, Zang Li
APWeb-WAIM 2024, Full Paper [\[Code\]](#)

Empowering Legal Citation Recommendation via Efficient Instruction-Tuning of Pre-trained Language Models

Jie Wang, Kanha Bansal, Ioannis Arapakis, Xuri Ge, Joemon M. Jose
ECIR 2024, Full Paper

Deep Memory Network with Bi-LSTM for Personalized Context-Aware Citation Recommendation

Jie Wang, Li Zhu, Tao Dai, Yabin Wang
Neurocomputing, vol. 410, pp. 103-113, 2020

Low-Rank and Sparse Matrix Factorization with Prior Relations for Recommender Systems

Jie Wang, Li Zhu, Tao Dai, Qiannan Xu, Tianyu Gao
Applied Intelligence, vol. 51, no. 6, pp. 3435-3449, 2021

SELECTED CO-AUTHORED PUBLICATIONS

IISAN: Efficiently Adapting Multimodal Representation for Sequential Recommendation with Decoupled PEFT

Junchen Fu, Xuri Ge, Xin Xin, Alexandros Karatzoglou, Ioannis Arapakis, **Jie Wang**, Joemon M. Jose
SIGIR 2024, Full Paper

Capturing the Spectrum of Social Media Conflict: A Novel Multi-Objective Classification Model

Oliver Warke, Joemon M. Jose, Jan Breitsohl, **Jie Wang**
ICTIR 2024, Full Paper

3SHNet: Boosting Image-Sentence Retrieval via Visual Semantic-Spatial Self-Highlighting

Xuri Ge, Songpei Xu, Fuhai Chen, **Jie Wang**, Guoxin Wang, Shan An, Joemon M. Jose
Information Processing & Management, 2024

Exploring Adapter-Based Transfer Learning for Recommender Systems: Empirical Studies and Practical Insights

Junchen Fu, Fajie Yuan, Yu Song, Zheng Yuan, Mingyue Cheng, Shenghui Cheng, Jiaqi Zhang, **Jie Wang**, Yunzhu Pan
WSDM 2024, Full Paper

NineRec: A Benchmark Dataset Suite for Evaluating Transferable Recommendation

Jiaqi Zhang, Yu Cheng, Yongxin Ni, Yunzhu Pan, Zheng Yuan, Junchen Fu, Youhua Li, **Jie Wang**, Fajie Yuan
IEEE TPAMI, Special Issue, 2024

OPEN-SOURCE RESEARCH

MODT4R

Multi-objective decision transformer for sequential recommendation, with accuracy, diversity, and novelty evaluation.

LEA

Research code accompanying the SIGIR 2024 LLM plus reinforcement-learning recommender.

TransRec

Transferable recommendation from multimodal user feedback and content representations.

HONORS AND AWARDS

WSDM Student Travel Grant, ACM	2025
Excellent Graduate, Xi'an Jiaotong University	2021
Excellent New Postgraduate Scholarship, Xi'an Jiaotong University	2018 - 2020
National Encouragement Scholarship, Dalian University of Technology	2016
Undergraduate Scholarship, Dalian University of Technology	2015 - 2017

TEACHING

Artificial Intelligence, Honors and Master's level	2024 - 2025
Machine Learning, Honors and Master's level	2024 - 2025
Master's Thesis Supervision	2023 - 2024
Introduction to Object-Oriented Programming, Level 2	2023 - 2024
Web Science, M.Sc. level	2022 - 2023; 2024 - 2025
Computing Science 1S - Computing Systems, Level 1	2022 - 2024

ACADEMIC ACTIVITIES

Presentations CIKM 2024 oral presentation, Boise, USA; WSDM 2025 presentation, Hannover, Germany.

Invited Lectures Machine Learning and Deep Learning Overview; Applications of ML and DL, January 2025.

Reviewer ICLR, WWW, SIGIR, CIKM, WSDM, AAAI, IEEE TPAMI, and related venues.

TECHNICAL SKILLS

Research: Recommender systems, reinforcement learning, large language models, multimodal retrieval, video understanding, diffusion models.

Engineering: Python, PyTorch, distributed model training, large-scale retrieval evaluation, experiment design, user studies.

Languages: Chinese (native), English (professional).

REFERENCES

Available upon request.